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Microfluidic devices with multiple inlets and outlets

Abstract

Techniques regarding nanofluidic chips with a plurality of inlets and/or outlets in fluid communication with one or more nanoDLD arrays are provided. For example, one or more embodiments described herein can comprise a nanoscale deterministic lateral displacement array between and in fluid communication with a global inlet and a global outlet. The nanoscale deterministic lateral displacement array can further be between and in fluid communication with a local inlet and a local outlet. Also, the nanoscale deterministic lateral displacement array can laterally displace a particle comprised within a sample fluid supplied from the global inlet to a collection region that directs the particle to the local outlet. An advantage of such an apparatus can be the expanded versatility of the nanoscale deterministic lateral displacement array for sample preparation applications involving nanoparticles not accessible to other higher throughput microscale microfluidic technologies.

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Background/Summary

BACKGROUND

(1) The subject disclosure relates to microfluidic devices comprising multiple inlets and/or outlets, and more specifically, to microfluidic chips comprising one or more nanoscale deterministic lateral displacement arrays in fluid communication with multiple inlets and/or outlets.

(2) The need for nanoscale particle separation, concentration, and purification from complex mixtures is ubiquitous in medicine, food safety, agriculture, and biotechnology; particularly with the growing role played by genomics in these fields. For example, deoxyribonucleic acid (“DNA”) and exosomes, small extracellular vesicles (EVs) containing a host of biomarkers (e.g. tumor-specific proteins, micro-ribonucleic acid (“microRNA”), messenger ribonucleic acid (“mRNA”), and DNA), can operate as potentially powerful prognosticators and/or therapeutic agents for diagnosing, monitoring, and/or treating disease. Length-based separation of DNA has primarily been served by established techniques including gel electrophoresis, high-performance liquid chromatography, precipitation, and/or affinity capture. Similarly, for exosomes, ultracentrifugation (“UC”), filtration, precipitation, and/or affinity capture can be applied to enact a separation or purification of extracellular vesicles (“EVs”). However, conventional methods suffer from one or more of the following issues: manual preparation required from a trained technician; batch rather than continuous processing; expensive equipment necessary for operation; sample loss; inability to handle small volumes; long run times; and/or background contamination.

(3) Microfluidic lab-on-a-chip (“LOC”) devices have drawn the gaze of many academic and/or industry leaders searching overcome problems with conventional devices. Microfluidic technologies can offer: fast analysis and response time; low fabrication costs (can be mass

produced and integrated into multiple processes such as labeling, purification, separation, and detection); high-throughput analysis; and/or low fluid volumes (can process with less waste, lower reagent costs when needed and/or less sample volume can be needed for a diagnostic). Within this framework, techniques employing a host of mechanisms have been explored and exploited for nanoparticle separation in general, such as field flow fractionation, centrifugal, optical, affinity capture, electrophoresis, dielectrophoresis, magnetophoresis, acoustophoresis, ion concentration polarization, electrohydrodynamic vortices, deterministic lateral displacement, and/or sieving

(4) While most of the features mentioned above for microfluidic technologies are overtly aligned with present needs, lower required fluid volumes appear to conflict with the goal of high throughput or shorter processing times when applied to sample preparation where sufficient volume is needed for downstream, off-chip analysis. It is this feature that reduces this list of usable microfluidic techniques as applied to sample preparation of nanoscale biomarkers; however, there are several of these methods that can reasonably employ high-throughput analysis to parallelize the separation process, effectively scaling the volume up.

(5) One technology that has proven versatile in its ability to separate a range of nanoscale particles with different morphology is nanoscale lateral deterministic displacement (“nanoDLD”). NanoDLD technology shrinks deterministic lateral displacement technology to the nanoscale, demonstrating the ability to sub-fractionate exosome populations (e.g., 30-150 nanometers (“nm”) in size) with tens of nanometers resolution in a continuous flow system (e.g., without batch processing). The technology can be used to demonstrate separation of DNA fragments by length with a selectivity of about 200 base pairs (“bp”) difference. A key challenge for nanoDLD is very low sample throughput per array (e.g., about 0.2 nanoliters per hour (“nL/hr”)) owing to the technology's scaled dimensions. By contrast, micron scale DLD pillar arrays can be less fluidically restrictive and used to isolate circulating tumor cells (“CTCs”) from blood at flow rates of 10 milliliters per minute (“mL/min”), or 600 mL/hr, for twinned arrays (e.g., 3 billion times faster than a single nanoDLD array. Of course, micron scale DLD arrays cannot separate nanoscale colloids, only much larger cellular material.

SUMMARY

(6) The following presents a summary to provide a basic understanding of one or more embodiments of the invention. This summary is not intended to identify key or critical elements, or delineate any scope of the particular embodiments or any scope of the claims. Its sole purpose is to present concepts in a simplified form as a prelude to the more detailed description that is presented later. In one or more embodiments described herein can comprise apparatuses, systems, and/or methods regarding microfluidic devices with one or more nanoDLD arrays in fluid communication with multiple inlets and/or outlets are described.

(7) According to an embodiment, an apparatus is provided. The apparatus can comprise a nanoscale deterministic lateral displacement array between and in fluid communication with a global inlet and a first outlet. The nanoscale deterministic lateral displacement array can further be between and in fluid communication with a local inlet and a second outlet. Also, the nanoscale deterministic lateral displacement array can laterally displace a particle comprised within a sample fluid supplied from the global inlet to a collection region that directs the particle to the second outlet. An advantage of such an apparatus can be the expanded versatility of the nanoscale deterministic lateral displacement array for sample preparation applications involving nanoparticles not accessible to other higher throughput microscale microfluidic technologies.

(8) In some examples, the apparatus can comprise a second local inlet in fluid communication with the nanoscale deterministic lateral displacement array. The global inlet, the local inlet, and the second local inlet can be located at a side of the nanoscale deterministic lateral displacement array opposite the first outlet and the second outlet. An advantage of such an apparatus can be the use of multiple fluids inputted via the multiple inlets and subject to manipulation by the nanoscale deterministic lateral displacement array.

(9) According to an embodiment, an apparatus is provided. The apparatus can comprise a layer of a microfluidic chip. The layer can comprise a nanoscale deterministic lateral displacement array positioned between and in fluid communication with a global inlet and a global outlet. The nanoscale deterministic lateral displacement array can further be positioned between and in fluid communication with a plurality of local inlets and a plurality of local outlets. The nanoscale deterministic lateral displacement array can laterally displace particles comprised within a sample fluid supplied from the global inlet to a plurality of collection regions. An advantage of such an apparatus can be the use of a microfluidic chip to perform high throughput operations using nanoscale deterministic lateral displacement technology.

(10) In some examples, a first side of the layer can comprise the nanoscale deterministic lateral displacement array. Also, a second side of the layer can comprise a first fluid reservoir in fluid communication with the plurality of local inlets and a second fluid reservoir in fluid communication with the plurality of local outlets. An advantage of such an apparatus can be the enablement of design layouts that can arrange one or more nanoscale deterministic lateral displacement arrays in geometric formations that can facilitate high density.

(11) According to an embodiment, a method is provided. The method can comprise directing, by a microfluidic chip, a sample fluid from a global inlet and through a nanoscale deterministic lateral displacement array. The method can also comprise directing, by the microfluidic chip, a second fluid through a collection region that extends from a local inlet, through the nanoscale deterministic lateral displacement array, and into an outlet. Further, the method can comprise displacing, by the nanoscale deterministic lateral displacement array, a sample comprised within the sample fluid in a lateral direction from the sample fluid and into the collection region. An advantage of such a method can be the use of a nanoscale deterministic lateral displacement technology with a microfluidic chip to facilitate one or more sample preparation processes.

(12) In some examples, the method can comprise directing, by the microfluidic chip, a third fluid from a second local inlet and through the nanoscale deterministic lateral displacement array to form a stream of third fluid. Also, the displacing the sample in the lateral direction can comprise displacing the sample through the stream of third fluid. An advantage of such a method can be the enablement of complex (e.g., multi-stage) sample preparation processes using nanoscale deterministic lateral displacement technology with a microfluidic chip.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates a diagram of example, non-limiting microfluidic channels that can be located on layer of a microfluidic chip in accordance with one or more embodiments described herein.

(2) FIG. 2 illustrates another diagram of example, non-limiting microfluidic channels that can be located on layer of a microfluidic chip in accordance with one or more embodiments described herein.

(3) FIG. 3 illustrates another diagram of example, non-limiting microfluidic channels that can be located on layer of a microfluidic chip in accordance with one or more embodiments described herein.

(4) FIG. 4 illustrates a diagram of an example, non-limiting first surface of a microfluidic chip layer that can comprise a plurality of nanoDLD arrays in fluid communication with a plurality of inlets and/or outlets in accordance with one or more embodiments described herein.

(5) FIG. 5 illustrates a diagram of an example, non-limiting second surface of a microfluidic chip layer that can comprise a plurality of fluid reservoirs arranged based on the alignment of one or more nanoDLD arrays positioned on another layer of the microfluidic chip in accordance with one or more embodiments described herein.

- (6) FIG. 6 illustrates a diagram of an example, non-limiting reservoir gasket that can comprise a surface of a microfluidic chip layer in accordance with one or more embodiments described herein.
- (7) FIG. 7 illustrates another diagram of an example, non-limiting reservoir gasket that can comprise a surface of a microfluidic chip layer in accordance with one or more embodiments described herein.
- (8) FIG. 8 illustrates another diagram of an example, non-limiting reservoir gasket that can comprise a surface of a microfluidic chip layer in accordance with one or more embodiments described herein.
- (9) FIG. 9 illustrates a flow diagram of an example, non-limiting method that can facilitate high throughput sample preparation and/or processing using one or more nanoDLD arrays in a microfluidic chip in accordance with one or more embodiments described herein.
- (10) FIG. 10 illustrates another flow diagram of an example, non-limiting method that can facilitate high throughput sample preparation and/or processing using one or more nanoDLD arrays in a microfluidic chip in accordance with one or more embodiments described herein.

DETAILED DESCRIPTION

- (11) The following detailed description is merely illustrative and is not intended to limit embodiments and/or application or uses of embodiments. Furthermore, there is no intention to be bound by any expressed or implied information presented in the preceding Background or Summary sections, or in the Detailed Description section.
- (12) One or more embodiments are now described with reference to the drawings, wherein like referenced numerals are used to refer to like elements throughout. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a more thorough understanding of the one or more embodiments. It is evident, however, in various cases, that the one or more embodiments can be practiced without these specific details.
- (13) Given the above problems with conventional sample analysis and/or preparation techniques and/or the limitations of conventional microfluidic devices; the present disclosure can be implemented to produce a solution to one or more of these problems in the form of a one or more nanoDLD arrays in fluid communication with a plurality of inlets and/or a plurality of outlets. Advantageously, the use of multiple inlets and/or outlets can facilitate complex sample preparation and/or analysis techniques. Further, various embodiments described herein can regard one or more microfluidic chips that can comprise a plurality of nanoDLD arrays, each is respective communication with a plurality of inlets and/or a plurality of outlets. Advantageously, the one or more microfluidic chips described herein can perform the complex sample preparation and/or analysis techniques with a throughput rate substantially greater than conventional microfluidic techniques that use nanoDLD technology.
- (14) As used herein, the term “lab-on-a-chip (“LOC”)” can refer to one or more devices that can integrate one or more laboratory functions onto an integrated circuit (e.g., a semiconductor substrate structure) to achieve autonomous preparation, screening, and/or analysis of one or more samples. LOCs can utilize microelectromechanical systems and/or microfluidic systems. One of ordinary skill in the art will recognize that a LOC devices can range in size from, for example, one or more square millimeters to one or more square centimeters.
- (15) As used herein the term “deterministic lateral displacement (“DLD”)” can refer to one or more microfluidic techniques that can size fractionate a polydisperse suspension of molecules using one or more arrays of obstacles. For example, DLD arrays can laterally displace target molecules within a sample stream based on size. Further, DLD arrays can comprise a plurality of pillars arranged in a lattice structure. Rows of pillars comprising the lattice structure can be positioned offset of each other at a defined angle, and pillars can be separated from each other by a defined gap size. The defined angle and/or gap size can facilitate displacement of one or more molecules of a target size range comprised within a stream flowing through the DLD array.
- (16) As used herein the term “nanoDLD array” can refer to a DLD array that can be characterized

by one or more dimensions ranging from greater than or equal to 1 nanometer (nm) and less than or equal to 999 nm. For example, a nanoDLD array can be a DLD array characterized by a gap size (e.g., a distance between adjacent pillars comprised within the lattice structure) of greater than or equal to 1 nm and less than or equal to 999 nm (e.g., greater than or equal to 25 nm and less than or equal to 250 nm).

(17) FIG. 1 illustrates a diagram of example, non-limiting microfluidic channels **100** that can comprise one or more nanoDLD arrays **102** in fluid communication with a plurality of inlets and/or outlets in accordance with one or more embodiments described herein. Repetitive description of like elements employed in other embodiments described herein is omitted for sake of brevity. The one or more microfluidic channels **100** can be delineated by dashed white lines in FIG. 1, which depicts three microfluidic channels **100** that can share a common nanoDLD array **102**. As shown in FIG. 1, the one or more microfluidic channels **100** can be positioned adjacent to each other. While FIG. 1 depicts three microfluidic channels **100** adjacent to each other, the architecture described herein is not so limited. One of ordinary skill in the art will readily recognize that fewer (e.g., two microfluidic channels **100**) or greater (e.g., four or more microfluidic channels **100**) microfluidic channels **100** than the three depicted in FIG. 1 can be positioned adjacent to each other and/or share a common nanoDLD array **102** based on one or more design preferences.

(18) The one or more nanoDLD arrays **102** (e.g., represented by a plurality of diagonal lines in FIG. 1) can comprise one or more lattices of asymmetric pillars arranged in rows and/or columns. FIG. 1 shows expanded views of multiple portions of the one or more nanoDLD arrays **102** to illustrate an exemplary structure. As shown in the expanded portion, the plurality of pillars comprised within the one or more nanoDLD arrays **102** can be arranged at an angle, such that one or more rows and/or columns of the pillars can be offset from adjacent rows and/or columns of the pillars. Further, the one or more nanoDLD arrays **102** can have a uniform gap size between pillars along the width (e.g., along the “X” direction) and/or length (e.g., along the “Y” direction) of the one or more microfluidic channels **100**. Alternatively, the one or more nanoDLD arrays **102** can have varying gap sizes between pillars along the width (e.g., along the “X” direction) and/or length (e.g., along the “Y” direction) of the one or more microfluidic channels **100**. For example, the gap size of the one or more nanoDLD arrays **102** can decrease (e.g., gradually and/or abruptly) along the length (e.g., along the “Y” direction) of the microfluidic channel **100**.

(19) In one or more embodiments, the one or more microfluidic channels **100** can comprise respective nanoDLD arrays **102**. For example, adjacent microfluidic channels **100** can comprise respective nanoDLD arrays **102**. Alternatively, in various embodiments one or more of the microfluidic channels **100** can share a common nanoDLD array **102** that can extend across the total width (e.g., along the “X” direction) of the microfluidic channels **100** (e.g., as depicted in FIG. 1). As shown in FIG. 1, the nanoDLD array **102** can be configured to periodically alternate the direction of lateral displacement (e.g., represented by the “LD” arrows shown in FIG. 1) exhibited by the nanoDLD array **102** so as to displace particles towards one or more collection regions **104** (e.g., delineated by black dashed lines in FIG. 1) located in the microfluidic channels **100**. For example, a nanoDLD array **102** shared by multiple microfluidic channels **100** can alternate the displacement direction at positions within the one or more collection regions **104** and between adjacent microfluidic channels **100** (e.g., as depicted by the expanded portions shown in the exemplary architecture of FIG. 1).

(20) The one or more microfluidic channels **100** can be in fluid communication with a global inlet **106** that can supply a sample fluid to the one or more microfluidic channels **100** via one or more inlet buses **108**. For example, the one or more inlet buses **108** can extend between the one or more microfluidic channels **100** to form one or more supply channels between adjacent microfluidic channels **100**. The sample fluid can be directed to the one or more nanoDLD arrays **102** via the one or more supply channels. The sample fluid can comprise one or more samples and/or one or more contaminants.

(21) The one or more microfluidic channels **100** can also be in fluid communication with a global outlet **110** that can receive the sample fluid downstream of the one or more nanoDLD arrays **102** via one or more outlet busses **112**. In various embodiments, the sample fluid can flow into the one or more nanoDLD arrays **102** via the one or more supply channels. While flowing through the one or more nanoDLD arrays **102**, one or more samples comprised in the sample fluid can be laterally displaced (e.g., “bumped”) towards a collection region **104** within a microfluidic channel **100**. Contaminants comprised within the sample fluid can flow through the one or more nanoDLD arrays **102** within minimal displacement (e.g., flow in a “zig zag” through the nanoDLD array **102**) and into the global outlet **110** via the one or more outlet buses **112**.

(22) Additionally, the one or more microfluidic channels **100** can comprise one or more local inlets **114** in fluid communication with the one or more nanoDLD arrays **102**. The one or more local inlets **114** can supply a second fluid to the one or more microfluidic channels **100**. For example, the second fluid can enter the one or more microfluidic channels **100** via the one or more local inlets **114** and flow into the one or more nanoDLD arrays **102**. For example, the second fluid can flow into the one or more collection regions **104**. In one or more embodiments, the second fluid can zig zag through the one or more nanoDLD arrays **102** with minimal lateral displacement. In various embodiments, the second fluid can be a buffer solution, which can facilitate purification of one or more samples laterally displaced into the collection region **104** by the one or more nanoDLD arrays **102**. Additionally, the one or more local inlets **114** can be in fluid communication with one or more fluid reservoirs containing the second fluid.

(23) Further, the one or more microfluidic channels **100** can comprise one or more local outlets **116** in fluid communication with the one or more nanoDLD arrays **102**. The one or more local outlets **116** can receive fluid flowing from the collection region **104** of the microfluidic channel **100**. For example, a microfluidic channel's **100** local outlet **116** can receive one or more samples laterally displaced into the collection region **104** from the sample fluid and/or the second fluid supplied to the collection region **104** by a local inlet **114**. Additionally, the one or more local outlets **116** can be in fluid communication with one or more fluid reservoirs. In one or more embodiments, the one or more microfluidic channels **100** can optionally comprise one or more filters positioned between the one or more local inlets **114** and/or the one or more nanoDLD arrays **102** and/or between the global inlet **106** and/or the one or more nanoDLD arrays **102**.

(24) In an exemplary use of the one or more microfluidic channels **100** depicted in FIG. 1, a complex biological sample (e.g., blood, urine, and/or a cultured media) can be inputted to the global inlet **106**. For instance, the biological sample can comprise desired sample particles such as exosomes. The biological sample can flow through the inlet bus **108** and into the one or more nanoDLD arrays **102** (e.g., via the one or more supply channels). As the biological sample flows through the one or more nanoDLD arrays **102**, the desired particles (e.g., exosomes of a defined size) can be laterally displaced to a collection region **104** of one or more adjacent microfluidic channels **100**. The lateral displacement can separate the desired particles from one or more smaller contaminants (e.g., proteins, lipids, and/or small molecules) in the biological sample that are too small to be laterally displaced by the one or more nanoDLD arrays **102**. Thus, the desired particles can be laterally displaced from the biological sample and into a collection region **104** while one or more contaminants in the biological sample can flow through the nanoDLD arrays **102** and into one or more outlet buses **112** for collection by a global outlet **110**.

(25) Further, in the exemplary use a buffer fluid can be inputted into the one or more microfluidic channels **100** via the one or more local inlets **114**. For example, the buffer fluid can be focus junction fed by a common buffer fluid reservoir defined by a gasket and in fluid communication with the one or more local inlets **114**. The buffer fluid can flow from the one or more local inlets **114** into the nanoDLD array **102**. For example, the buffer fluid can flow into the microfluidic channel's **100** collection region **104** located within the nanoDLD array **102**. In one or more embodiments, the buffer fluid can comprise molecules too small to be laterally displaced by the one

or more nanoDLD arrays **102** and thereby can flow through the nanoDLD arrays **102** with minimal displacement (e.g., can zig zag through the one or more nanoDLD arrays **102**). Thereby, the buffer fluid can create a buffer stream flowing through the collection region **104**, wherein the desired particles can be laterally displaced (e.g., bumped) into the buffer stream by the one or more nanoDLD arrays **102**. The buffer stream comprising the displaced desired particles can flow through the collection region **104**, exit the one or more nanoDLD arrays **102**, and be collected by the one or more local outlets **116**. The one or more local outlets **116** can direct the desired particles of the biological sample into one or more first reservoirs (e.g., one or more sample reservoirs), while the global outlet **110** can direct the contaminants of the biological sample to one or more second reservoirs (e.g., one or more waste reservoirs).

(26) While FIG. **1** depicts the one or more microfluidic channels **100** comprising a single local inlet **114** and/or a single local outlet **116**; however, the architecture of the one or more microfluidic channels **100** is not so limited. For example, a microfluidic channel **100** comprising multiple local inlets **114** and/or multiple local outlets **116** is also envisaged. Additionally, while FIG. **1** depicts one or more collection regions **104** extending to one or more local outlets **116**, the architecture of the one or more microfluidic channels **100** is not so limited. For example, in one or more embodiments the one or more local outlets **116** can be repositioned laterally (e.g., along the “X” direction) so as to collect particles that zig zag through the one or more nanoDLD arrays **102** (e.g., waste particles, such as contaminants in the sample fluid), while the one or more collection regions **104** can direct bumped particles (e.g., desired particles) to the global outlet **110**. In other words, the one or more collection regions **104** can extend to one or more local outlets **116** (e.g., as shown in FIG. **1**) or the global outlet **110** depending on the alignment of the one or more local outlets **116**. Further, the alignment of the one or more local outlets **116** can depend on whether a designer prefers the desired particles to be collected by one or more local outlets **116** or a global outlet **110**.

(27) FIG. **2** illustrates a diagram of the example, non-limiting microfluidic channels **100** further comprising one or more second local inlets **202** in accordance with one or more embodiments described herein. Repetitive description of like elements employed in other embodiments described herein is omitted for sake of brevity. As shown in FIG. **2**, the one or more second local inlets **202** can also be in fluid communication with the one or more nanoDLD arrays **102**.

(28) The one or more second local inlets **202** can supply one or more third fluids to the one or more microfluidic channels **100**. The one or more third fluids can flow from the one or more second local inlets **202**, through the one or more nanoDLD arrays **102**, and into the global outlet **110** (e.g., via the one or more outlet buses **112**). The one or more second local inlets **202** can be in fluid communication with one or more fluid reservoirs comprising the third fluid. Also, as shown in FIG. **2**, the third fluid supplied to the microfluidic channel **100** by the one or more second local inlets **202** can be introduced into one or more regions of the one or more nanoDLD arrays **102** that are adjacent to the collection region **104**. In some embodiments, the third fluid can flow through the one or more nanoDLD arrays **102** with minimal displacement, thereby forming one or more streams flowing adjacent to the respective collection regions **104**. The streams of the third fluid can flow outside the collection region **104** and thereby flow into the one or more outlet buses **112** rather than the one or more local outlets **116**.

(29) In an exemplary use of the one or more microfluidic channels **100** depicted in FIG. **2**, a complex biological sample (e.g., blood, urine, and/or a cultured media) can be inputted to the global inlet **106**. For instance, the biological sample can comprise desired sample particles such as exosomes. The biological sample can flow through the inlet bus **108** and into the one or more nanoDLD arrays **102** (e.g., via the one or more supply channels). As the biological sample flows through the one or more nanoDLD arrays **102**, the desired particles (e.g., exosomes of a defined size) can be laterally displaced to a collection region **104** of one or more adjacent microfluidic channels **100**. The lateral displacement can separate the desired particles from one or more smaller contaminants (e.g., proteins, lipids, and/or small molecules) in the biological sample that are too

small to be laterally displaced by the one or more nanoDLD arrays **102**. Thus, the desired particles can be laterally displaced from the biological sample and into a collection region **104** while one or more contaminants in the biological sample can flow through the nanoDLD arrays **102** and into one or more outlet buses **112** for collection by a global outlet **110**.

(30) Additionally, in the exemplary use a labeling fluid can be inputted into the one or more microfluidic channels **100** via the one or more second local inlets **202**. For example, the labeling fluid can comprise one or more molecules having an affinity to couple with the one or more desired particles. The one or more molecules can be readily identifiable optically, electrochemically, and/or magnetically; thereby rendering the desired particles readily identifiable when coupled with the labeling fluid. For instance, the labeling fluid can facilitate one or more optical labeling techniques including, but not limited to: enzymatic labeling, chemical labeling, immunochemical labeling, protein labeling, genetic labeling, DNA intercalating agents, a combination thereof and/or the like. The labeling fluid can be focus junction fed by a common labeling fluid reservoir defined by a gasket and in fluid communication with the one or more second local inlets **202**.

(31) The labeling fluid can flow from the one or more second local inlets **202** into the one or more nanoDLD arrays **102**. In one or more embodiments, the labeling fluid can comprise labeling molecules too small to be laterally displaced by the one or more nanoDLD arrays **102** and thereby can flow through the nanoDLD arrays **102** with minimal displacement (e.g., can zig zag through the one or more nanoDLD arrays **102**). Thereby, the labeling fluid can create one or more labeling streams flowing through the one or more nanoDLD arrays **102** (e.g., adjacent to the collection region **104**). The desired particles can be laterally displaced (e.g., bumped) into one or more of the labeling streams by the one or more nanoDLD arrays **102** while being directed, by the one or more nanoDLD arrays **102**, into a collection region **104**. Labeling molecules of the labeling fluid that do not couple with one or more desired particles can remain in the one or more labeling streams and/or flow into the one or more outlet buses **112**. In contrast, labeling molecules that couple with one or more desired particles being laterally displaced through the one or more labeling streams can be displaced with the one or more desired particles into the one or more collection regions **104**.

(32) Further, in the exemplary use a buffer fluid can be inputted into the one or more microfluidic channels **100** via the one or more local inlets **114**. For example, the buffer fluid can be focus junction fed by a common buffer fluid reservoir defined by a gasket and in fluid communication with the one or more local inlets **114**. The buffer fluid can flow from the one or more local inlets **114** into the nanoDLD array **102**. For example, the buffer fluid can flow into the microfluidic channel's **100** collection region **104** located within the nanoDLD array **102**. In one or more embodiments, the buffer fluid can comprise molecules too small to be laterally displaced by the one or more nanoDLD arrays **102** and thereby can flow through the nanoDLD arrays **102** with minimal displacement (e.g., can zig zag through the one or more nanoDLD arrays **102**). Thereby, the buffer fluid can create a buffer stream flowing through the collection region **104**, wherein the labeled desired particles (e.g., desired particles coupled with one or more labeling molecules due to an interaction with one or more labeling streams during lateral displacement) can be laterally displaced (e.g., bumped) into the buffer stream by the one or more nanoDLD arrays **102**. The buffer stream comprising the displaced and labelled desired particles can flow through the collection region **104**, exit the one or more nanoDLD arrays **102**, and be collected by the one or more local outlets **116**. The one or more local outlets **116** can direct the labeled desired particles of the biological sample into one or more first reservoirs (e.g., one or more sample reservoirs), while the global outlet **110** can direct the contaminants of the biological sample to one or more second reservoirs (e.g., one or more waste reservoirs). Therefore, the configuration depicted in FIG. 2 can facilitate purifying and labeling one or more samples within a single microfluidic channel **100** using one or more nanoDLD arrays **102**.

(33) FIG. 3 illustrates a diagram of the example, non-limiting microfluidic channels **100** further comprising one or more second local outlets **302** in accordance with one or more embodiments

described herein. Repetitive description of like elements employed in other embodiments described herein is omitted for sake of brevity. As shown in FIG. 3 the one or more second local outlets **302** can also be in fluid communication with the one or more nanoDLD arrays **102**.

(34) As shown in FIG. 3, the one or more collection regions **104** can direct fluids into both the one or more local outlets **116** and/or the one or more second local outlets **302**. For example, fluids exiting from a central portion of the one or more collection regions **104** can be collected by the one or more local outlets **116**, while fluids exiting from perimeter portions of the one or more collection regions **104** can be collected by the one or more second local outlets **302**. For instance, the one or more nanoDLD arrays **102** can laterally displace particles to different positions within the one or more collection regions **104** based on the size of the particles and/or the geometry of the one or more nanoDLD arrays **102**. For example, the one or more nanoDLD arrays **102** can laterally displace particles of a first size into a central portion of the one or more collection regions **104**, while laterally displacing particles of a second size (e.g., a size smaller than the first size) into a periphery portion (e.g., along the perimeter) of the one or more collection regions **104**. For instance, the gap size of the one or more nanoDLD arrays **102** can vary along the width (e.g., along the “X” direction) of the one or more microfluidic channels **100** to facilitate one or more sub-fractionations of particles displaced from the stream of sample fluid.

(35) Therefore, the one or more nanoDLD arrays **102** can fractionate particles from the sample fluid into the one or more collection regions **104** and/or sub-fractionate particles in the one or more collection regions **104** into various portions of the one or more collection regions **104**. Further, fractionated particles from the sample fluid can be collected by either the one or more local outlets **116** or the one or more second local outlets **302** based on the sub-fractionation occurring within the one or more collection regions **104**. For example, molecules of the sample fluid that are not fractionated by the one or more nanoDLD arrays **102** can be collected by the global outlet **110**, particles fractionated from the sample fluid by the one or more nanoDLD arrays **102** can be collected by the one or more second local outlets **302**, and/or particles sub-fractionated, by the one or more nanoDLD arrays **102**, from the fractionated particles can be collected by the one or more local outlets **116**.

(36) The microfluidic channel **100** configurations depicted and/or described herein with regards to FIGS. 2 and/or 3 can exemplify that the one or more microfluidic channels **100** can comprise a plurality of localized inlets and/or a plurality of localized outlets. The architecture of one or more microfluidic channels **100** is not limited to the number of localized inlets and/or the number of localized outlets depicted in the configurations of FIGS. 2 and 3. For example, the one or more microfluidic channels **100** can comprise more localized inlets than the two depicted in FIG. 2 (e.g., local inlet **114** and/or second local inlet **202**). One of ordinary skill in the art will readily recognize that the features and/or principals described herein to introduce an additional localized inlet (e.g., second local inlet **202**) to the architecture of the one or more microfluidic channels **100** can be expanded and/or repeated to introduce further localized inlets (e.g., configurations comprising local inlet **114**, second local inlet **202**, and/or a third local inlet). Further, each additional localized inlet can supply a respective fluid into the one or more microfluidic channels **100** to facilitate additional preparation, screening, and/or analysis operations. In another example, the one or more microfluidic channels **100** can comprise more localized outlets than the two depicted in FIG. 3 (e.g., local outlet **116** and/or second local outlet **302**). One of ordinary skill in the art will readily recognize that the features and/or principals described herein to introduce an additional localized outlet (e.g., second local outlet **302**) to the architecture of the one or more microfluidic channels **100** can be expanded and/or repeated to introduce further localized outlets (e.g., configurations comprising local outlet **116**, second local outlet **302**, and/or a third local outlet). Further, each additional localized outlet can collect a respective size group of fractionated particles from the sample fluid.

(37) Additionally, configurations of the one or more microfluidic channels **100** that combine one or

more of the various features described herein are also envisaged. For example, a configuration of the one or more microfluidic channels **100** comprising the one or more second local inlets **202** and the one or more second local outlets **302** is also envisaged.

(38) FIG. **4** illustrates a diagram of an example, non-limiting first side **400** of a microfluidic chip layer that can comprise a plurality of microfluidic channels **100** in accordance with one or more embodiments described herein. Repetitive description of like elements employed in other embodiments described herein is omitted for sake of brevity. As shown in FIG. **4**, a plurality of the microfluidic channels **100** described herein can be arranged on the first side **400** of a microfluidic chip layer to facilitate a high throughput of a sample fluid.

(39) The first side **400** can comprise a plurality of nanoDLD arrays **102**, wherein each nanoDLD array **102** can be comprised within a plurality of microfluidic channels **100**. For example, FIG. **4** depicts a first side **400** comprising four nanoDLD arrays **102** (e.g., one on each side of the first side **400**). Further, an expanded portion of the first side **400** delineates that the each of the nanoDLD arrays **102** can be shared by a plurality of microfluidic channels **100** arranged in accordance with one or more embodiments described herein.

(40) The plurality of nanoDLD arrays **102**, and/or thereby the plurality of microfluidic channels **100**, can be in fluid communication with a common global inlet **106** via an inlet bus **108** that can extend around the perimeter of the first side **400** and/or reach each of the nanoDLD arrays **102** and/or microfluidic channels **100**. Similarly, the plurality of nanoDLD arrays **102**, and/or thereby the plurality of microfluidic channels **100**, can be in fluid communication with a common global outlet **110** via an outlet bus **112** that can extend to each of the nanoDLD arrays **102** and/or microfluidic channels **100**.

(41) The expanded portion of FIG. **4** shows an exemplary configuration of the plurality of microfluidic channels **100**. While FIG. **4** depicts a first side **400** comprising multiple microfluidic channels **100** having the configuration depicted and/or described with regards to FIG. **1**, the architecture of the first side **400** is not so limited. As described herein, one or more of the plurality of microfluidic channels **100** can comprise a plurality of localized inlets and/or a plurality of localized outlets. For example, one or more of the plurality of microfluidic channels **100** can have the exemplary configurations depicted and/or described with regards to FIGS. **2** and/or **3**. Further, in some embodiments, a plurality of microfluidic channels **100** sharing a first nanoDLD array **102** on the first side **400** can have a different configuration (e.g., a different number of localized inlets and/or outlets) than a plurality of microfluidic channels **100** sharing a second nanoDLD array **102** on the first side **400**.

(42) FIG. **5** illustrates a diagram of an example, non-limiting second side **500** of the microfluidic chip layer that can comprise a plurality of reservoirs to facilitate supplying fluid to and/or collecting fluid from the plurality of microfluidic channels **100** comprised on the first layer **400** in accordance with one or more embodiments described herein. Repetitive description of like elements employed in other embodiments described herein is omitted for sake of brevity. In various embodiments, the first side **400** can be a front side of the microfluidic layer, while the second side can be a backside of the same microfluid layer. In one or more embodiments, the second side **500** can be comprised on a microfluidic chip layer that is separate and/or distinct from a microfluidic chip layer comprising the first side **400**. Also, the configuration of the second side **500** can correspond to the configuration of microfluidic channels **100** and/or nanoDLD arrays **102** positioned on the first side **400**. For instance, the black dots depicted in FIG. **5** can be presented solely for clarification purposes and/or can represent where the plurality of localized inlets (e.g., the plurality of local inlets **114**) and/or localized outlets (e.g., the plurality of local outlets **116**) can align with the second side.

(43) In one or more embodiments the global inlet **106** and/or the global outlet **110** can extend through the first side **400** and/or the second side **500** of the microfluidic chip (e.g., as shown in FIG. **5**). Additionally, in various embodiments, the second side **500** can comprise one or more seals

502 (e.g., O-rings) that can define a plurality of fluid reservoirs. For example, the one or more seals **502** can define a first fluid reservoir **504** that can be aligned with the plurality of localized inlets (e.g., plurality of local inlets **114**) comprising the multiple microfluidic channels **100** of the first side **400**. For instance, the plurality of black dots shown in the first fluid reservoir **504** in FIG. 5 can represent where the plurality of localized inlets from the microfluidic channels **100** of the first side **400** would align. Additionally, the one or more seals **502** can define a second fluid reservoir **506** that can be aligned with the plurality of localized outlets (e.g., plurality of local outlets **116**) comprising the multiple microfluidic channels **100** of the first side **400**. For instance, the plurality of black dots shown in the second fluid reservoir **506** **504** in FIG. 5 can represent where the plurality of localized outlets from the microfluidic channels **100** of the first side **400** would align.

(44) In various embodiments, the first fluid reservoir **504** can be in fluid communication with plurality of microfluidic channels **100** of the first side **400** via the plurality of localized inlets (e.g., the plurality of local inlets **114**). For example, the first fluid reservoir **504** can contain a fluid (e.g., a buffer fluid) that can be supplied to the plurality of microfluidic channels **100** via the one or more localized inlets. Also, the second fluid reservoir **506** can be in fluid communication with plurality of microfluidic channels **100** of the first side **400** via the plurality of localized outlets (e.g., the plurality of local outlets **116**). For example, the second fluid reservoir **506** can contain fluid (e.g., one or more samples) collected by the plurality of localized outlets of the microfluidic channels **100** of the first side **400**. Further, the plurality of fluid reservoirs can comprise one or more respective access vias (not shown) to facilitate supplying fluid to the fluid reservoir and/or collecting fluid from the fluid reservoir.

(45) While FIG. 5 depicts the second side **500** comprising two fluid reservoirs (e.g., first fluid reservoir **504** and/or second fluid reservoir **506**), the architecture of the second side **500** is not so limited. For example, the second side **500** can comprise additional fluid reservoirs based on the number of localized inlets (e.g., local inlets **114**) and/or localized outlets (e.g., local outlets **116**) comprised within the plurality of microfluidic channels **100** positioned on the first side **400**. For instance, wherein the plurality of microfluidic channels **100** on an adjacent first side **400** comprise second local inlets **202**, the second side **500** can comprise one or more additional fluid reservoir (e.g., defined by the one or more seals **502**) aligned and in fluid communication with the plurality of second local inlets **202** so as to supply a fluid to the microfluidic channels **100** via the one or more second local inlets **202**. In another instance, wherein the plurality of microfluidic channels **100** on an adjacent first side **400** comprise second local outlets **302**, the second side **500** can comprise one or more additional fluid reservoir (e.g., defined by the one or more seals **502**) aligned and in fluid communication with the plurality of second local outlets **302** so as to facilitate collection of sub-fractionated groups of sample from the plurality of microfluidic channels **100**.

(46) FIG. 6 illustrates a diagram of an example, non-limiting reservoir gasket **602** that can comprise the one or more second sides **500** of the one or more microfluidic chips in accordance with one or more embodiments described herein. Repetitive description of like elements employed in other embodiments described herein is omitted for sake of brevity. In various embodiments, the plurality of reservoirs comprising one or more second sides **500** can be defined by one or more reservoir gaskets **602** in addition to and/or alternatively to the one or more seals **502** described herein. For example, the one or more reservoir gaskets **602** can comprise a gasket with one or more voids that can define the plurality of reservoirs. For instance, a first void in the reservoir gasket **602** can define the first fluid reservoir **504**, and a second void in the reservoir gasket **604** can define the second fluid reservoir **506**. Additionally, the reservoir gasket **602** can comprise further voids to enable the global inlet **106** and/or the global outlet **110** to traverse through the reservoir gasket **602**.

(47) Example materials that can comprise the one or more reservoir gaskets **602** can include, but are not limited to: polymers (e.g., elastomers), thermosets, thermoplasts, super-molecular polymers, block copolymers, a combination thereof, and/or the like. In various embodiments, the one or more reservoir gaskets **602** can be created from one or more manufacturing processes, including, but not

limited to: three-dimensional printing, stamping, embossing, injection molding, laser cutting, ablating from a polymeric starting material, additive machining (e.g., laser-induced ink printing, polymerization, stereolithography, and/or three-dimensional extrusion printing), a combination thereof, and/or the like.

(48) FIG. 6 also depicts an exemplary first side **400** that can be adjacent to the exemplary second side **500**. Presenting the first side **400** next to the second side **500** can exemplify that the position, number, and/or configuration of the plurality of reservoirs of the second side **500** can be based on the position, number, and/or configuration of the microfluidic channels **100** comprised on the first side **400**.

(49) In one or more embodiments, a plurality of microfluidic channels **100** can be positioned on the first side **400** in various configurations to facilitate a high density of microfluidic channels **100** in the microfluidic chip. Advantageously, by utilizing many microfluidic channels **100**, the microfluidic chips described herein can implement nanoscale DLD technology while achieving higher throughput rates than conventional techniques. Further, the second side **500** can enable simultaneous fluid loading of all inlets associated with a particular fluid. For example, the plurality of local inlets **114** on the first side **400** can be simultaneously supplied fluid (e.g., a buffer fluid) from a fluid reservoir (e.g., the first fluid reservoir **504**) on the second side **500**. In another example, while the plurality of local inlets **114** are being supplied fluid, a plurality of second local inlets **202** on the first side **400** can be simultaneously supplied a second fluid (e.g., a labeling fluid) from another fluid reservoir on the second side **500**. By supplying fluid to multiple microfluidic channels **100** via a common reservoir, the microfluidic chip can minimize the number of required access vias needed to load the subject fluid (e.g., each fluid reservoir can have one respective access via). Similarly, by collecting samples from multiple microfluidic channels **100** in a common reservoir, the microfluidic chip can minimize the number of required access vias needed to extract the subject samples.

(50) One of ordinary skill in the art will readily recognize that the microfluidic channels **100** and/or the features of the first side **400** and/or second side **500** described herein can be implemented in a microfluidic chip in a variety of configurations to facilitate one or more desired performance characteristics (e.g., a desired throughput rate). For example, while FIGS. 4 and/or 6 depict a nanoDLD array **102** configuration comprising four nanoDLD arrays **102** positioned on respective sides of the first side **400**, additional configurations of nanoDLD arrays **102** are also envisaged. Additionally, fluid reservoir configurations based on the configuration of nanoDLD arrays **102** on an adjacent first side **400** are also envisaged. For example, the first side **400** can comprise a plurality of nanoDLD arrays **102** arranged in various geometric configurations, which can include, but are not limited to: square configurations, circular configurations, Greek spiral configurations, serpentine configurations, cross configurations (e.g., in the shape of an “x” and/or “+”), interdigitated configurations, tree configurations, convergent and/or divergent fractal network configurations, space-filled raster configurations, a combination thereof, and/or the like. Likewise, the second side **500** can comprise a plurality of fluid reservoirs arranged in various geometric configurations, which can include, but are not limited to: square configurations, circular configurations, Greek spiral configurations, serpentine configurations, cross configurations (e.g., in the shape of an “x” and/or “+”), interdigitated configurations, tree configurations, convergent and/or divergent fractal network configurations, space-filled raster configurations, a combination thereof, and/or the like.

(51) FIG. 7 illustrates a diagram of an example, non-limiting Greek spiral configuration of nanoDLD arrays **102** on the first side **400** and/or a corresponding Greek spiral configuration of fluid reservoirs (e.g., a first fluid reservoir **504** and/or a second fluid reservoir **506**) on the second side **500**. Repetitive description of like elements employed in other embodiments described herein is omitted for sake of brevity. As shown in FIG. 7, the plurality of nanoDLD arrays **102** and/or the plurality of fluid reservoirs described herein can be arranged in one or more configurations to

facilitate increased density in the microfluidic chip, such as a Greek spiral configuration.

(52) FIG. 8 illustrates a diagram of an example, non-limiting serpentine configuration of nanoDLD arrays **102** on the first side **400** and/or a corresponding serpentine configuration of fluid reservoirs (e.g., a first fluid reservoir **504** and/or a second fluid reservoir **506**) on the second side **500**.

Repetitive description of like elements employed in other embodiments described herein is omitted for sake of brevity. As shown in FIG. 8, the plurality of nanoDLD arrays **102** and/or the plurality of fluid reservoirs described herein can be arranged in one or more configurations to facilitate increased density in the microfluidic chip, such as a serpentine configuration.

(53) FIG. 9 illustrates a flow diagram of an example, non-limiting method **900** that can facilitate sample preparation, analysis, and/or screening using one or more microfluidic channels **100** that can comprise one or more nanoDLD arrays **102** in accordance with one or more embodiments described herein. Repetitive description of like elements employed in other embodiments described herein is omitted for sake of brevity.

(54) At **902**, the method **900** can comprise directing, by a microfluidic chip (e.g., via the first side **400** of the microfluidic chip), a sample fluid from a global inlet **106** and through one or more nanoDLD arrays **102** (e.g., into a global outlet **110**). For example, the sample fluid can be a complex biological sample including, but not limited to: blood, urine, a cultured media, saliva, cerebral spinal fluid, plasma, serum, feces, lavage, sputum, tears, sweat, a combination thereof, and/or the like. As described herein, the one or more nanoDLD arrays **102** can be configured to laterally displace particles in one or more directions based on size. In various embodiments, the sample fluid can be directed to the one or more nanoDLD arrays **102** via one or more inlet buses **108** and can be directed from the one or more nanoDLD arrays **102** via one or more outlet buses **112**.

(55) At **904**, the method **900** can comprise, directing, by the microfluidic chip (e.g., via the first side **400**), a second fluid through one or more collection regions **104** that can extend from one or more local inlets **114**, through the one or more nanoDLD arrays **102**, and/or into one or more outlets (e.g., one or more local outlets **116**). For example, the second fluid can be a fluid used to facilitate a preparation and/or modification process of particles within the one or more collection regions **104**. For instance, the second fluid can be a buffer fluid. In various embodiments, the one or more local inlets **114** can be in fluid communication with one or more fluid reservoirs (e.g., one or more first fluid reservoirs **504**) containing the second fluid on an adjacent side of the microfluidic chip (e.g., a second side **500**).

(56) At **906**, the method **900** can comprise, displacing, by the one or more nanoDLD arrays **102**, one or more samples comprised within the sample fluid in one or more lateral directions from the sample fluid and/or into the one or more collection regions **104**. For example, the sample fluid can be directed to a portion of the one or more nanoDLD arrays **102** located between microfluidic channels **100**, wherein the one or more nanoDLD arrays **102** can laterally displace the sample into a collection region **104** of one of the microfluidic channels **100**. Additionally, the sample can flow through the one or more collection regions **104** into the one or more local outlets **116** along with the second fluid (e.g., a buffer fluid). In various embodiments, the one or more local outlets **116** can be in fluid communication with one or more fluid reservoirs (e.g., one or more second fluid reservoirs **506**) that can collect the one or more samples on an adjacent side of the microfluidic chip (e.g., a second side **500**). Advantageously, the method **900** can use of nanoscale DLD technology with a microfluidic chip to facilitate one or more sample preparation processes.

(57) FIG. 10 illustrates a flow diagram of an example, non-limiting method **1000** that can facilitate sample preparation, analysis, and/or screening using one or more microfluidic channels **100** that can comprise one or more nanoDLD arrays **102** in accordance with one or more embodiments described herein. Repetitive description of like elements employed in other embodiments described herein is omitted for sake of brevity.

(58) At **1002**, the method **1000** can comprise directing, by a microfluidic chip (e.g., via the first

side **400** of the microfluidic chip), a sample fluid from a global inlet **106** and/or through one or more nanoDLD arrays **102** (e.g., into a global outlet **110**). For example, the sample fluid can be a complex biological sample including, but not limited to: blood, urine, a cultured media, saliva, cerebral spinal fluid, plasma, serum, feces, lavage, sputum, tears, sweat, a combination thereof, and/or the like. As described herein, the one or more nanoDLD arrays **102** can be configured to laterally displace particles in one or more directions based on size. In various embodiments, the sample fluid can be directed to the one or more nanoDLD arrays **102** via one or more inlet buses **108** and can be directed from the one or more nanoDLD arrays **102** via one or more outlet buses **112**.

(59) At **1004**, the method **1000** can comprise, directing, by the microfluidic chip (e.g., via the first side **400**), a second fluid through one or more collection regions **104** that can extend from one or more local inlets **114**, through the one or more nanoDLD arrays **102**, and/or into one or more outlets (e.g., one or more local outlets **116**). For example, the second fluid can be a fluid used to facilitate a preparation and/or modification process of particles within the one or more collection regions **104**. For instance, the second fluid can be a buffer fluid. In various embodiments, the one or more local inlets **114** can be in fluid communication with one or more fluid reservoirs (e.g., one or more first fluid reservoirs **504**) containing the second fluid on an adjacent side of the microfluidic chip (e.g., a second side **500**).

(60) At **1006**, the method **1000** can comprise, directing, by the microfluidic chip (e.g., via the first side **400**), a third fluid from one or more second local inlets **202** and/or through the one or more nanoDLD arrays **102** (e.g., and/or into the global outlet **110**) to form one or more streams of third fluid. For example, the third fluid can be a fluid used to facilitate a preparation and/or modification process of particles entering one or more collection regions **104**. For instance, the third fluid can be a labeling fluid in accordance with one or more embodiments described herein. In various embodiments, the one or more second local inlets **202** can be in fluid communication with one or more fluid reservoirs containing the third fluid on an adjacent side of the microfluidic chip (e.g., a second side **500**).

(61) At **1008**, the method **1000** can comprise, displacing, by the one or more nanoDLD arrays **102**, one or more samples comprised within the sample fluid in one or more lateral directions from the sample fluid and/or into the one or more collection regions **104**. For example, the sample fluid can be directed to a portion of the one or more nanoDLD arrays **102** located between microfluidic channels **100**, wherein the one or more nanoDLD arrays **102** can laterally displace the sample into a collection region **104** of one of the microfluidic channels **100**. The displacing at **1008** can comprise displacing the sample through the one or more streams of third fluid. For example, the one or more streams of third fluid can flow adjacent to the one or more collection regions **104**, wherein the one or more samples can pass through the one or more streams of third fluid to enter the one or more collection regions **104**. Additionally, the one or more samples can flow through the one or more collection regions **104** into the one or more local outlets **116** along with the second fluid (e.g., a buffer fluid). In various embodiments, the one or more local outlets **116** can be in fluid communication with one or more fluid reservoirs (e.g., one or more second fluid reservoirs **506**) that can collect the one or more samples on an adjacent side of the microfluidic chip (e.g., a second side **500**). Advantageously, the method **1000** can utilize a plurality of localized inlets (e.g., one or more local inlets **114** and/or one or more second local inlets **202**) to enable complex (e.g., multi-stage) sample preparation processes using nanoscale DLD technology with a microfluidic chip.

(62) In addition, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or.” That is, unless specified otherwise, or clear from context, “X employs A or B” is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. Moreover, articles “a” and “an” as used in the subject specification and annexed drawings should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to

a singular form. As used herein, the terms “example” and/or “exemplary” are utilized to mean serving as an example, instance, or illustration. For the avoidance of doubt, the subject matter disclosed herein is not limited by such examples. In addition, any aspect or design described herein as an “example” and/or “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects or designs, nor is it meant to preclude equivalent exemplary structures and techniques known to those of ordinary skill in the art.

(63) It is, of course, not possible to describe every conceivable combination of components, products and/or methods for purposes of describing this disclosure, but one of ordinary skill in the art can recognize that many further combinations and permutations of this disclosure are possible. Furthermore, to the extent that the terms “includes,” “has,” “possesses,” and the like are used in the detailed description, claims, appendices and drawings such terms are intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim. The descriptions of the various embodiments have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. An apparatus, comprising: a nanoscale deterministic lateral displacement array comprising alternating rows of inward displacement pillars and outward displacement pillars, wherein the alternating rows are connected and formed between and in fluid communication with an inlet bus and an outlet bus, and wherein the alternating rows form a Greek spiral or a serpentine configuration; a global inlet in fluid communication with the inlet bus; and a global outlet the outlet bus.
2. The apparatus of claim 1, wherein the nanoscale deterministic lateral displacement array further laterally displaces contaminate particles comprised within a sample fluid supplied from the inlet bus via the global inlet to the outlet bus and the global outlet, wherein the contaminate particles are smaller than particles of interest.
3. The apparatus of claim 1, wherein the nanoscale deterministic lateral displacement array laterally displaces particles of interest comprised within a sample fluid supplied from the inlet bus via the global inlet into streams of buffer fluid supplied from first local inlets that direct the particles of interest to first local outlets, and the apparatus further comprising: second local inlets formed on a first side of the alternating rows that receive and supply an additional fluid into the streams of buffer fluid.
4. The apparatus of claim 3, wherein the additional fluid comprises a labeling fluid.
5. The apparatus of claim 4, wherein the particles of interest are labeled by the labeling fluid while being laterally displaced into the streams of buffer fluid, and wherein the particles of interest are purified by the buffer fluid from smaller contaminants comprised within the sample fluid.
6. The apparatus of claim 3, wherein the apparatus further comprises: second local outlets formed on a second side of the alternating rows.
7. The apparatus of claim 6, wherein the particles of interest comprise first particles of interest and wherein the nanoscale deterministic lateral displacement array laterally displaces second particles of interest comprised within the sample fluid to the second outlets, and wherein the second particles of interest are smaller than the first particles of interest.
8. The apparatus of claim 1, wherein the inlet bus is formed on a first side of the alternating rows and wherein the outlet bus is formed on a second side of the alternating rows.

9. The apparatus of claim 2, further comprising: collection regions formed within inward deflecting pillars, and wherein first local inlets and first local outlets are respectively adjacent to opposite sides of the collection regions.
10. The apparatus of claim 2, wherein the particles of interest comprise exosomes.
11. A microfluidic chip, comprising: a reservoir layer; and a nanoscale deterministic lateral displacement (nanoDLD) layer in fluid in communication with the reservoir layer and comprising: alternating rows of inward displacement pillars and outward displacement pillars, wherein the alternating rows are connected and formed between and in fluid communication with an inlet bus and an outlet bus; first local inlets formed on a first side of the alternating rows; and first local outlets formed on a second side of the alternating rows opposite the first side, wherein the nanoDLD layer comprises nanoDLD arrays of a first configuration on a first side of the microfluidic chip, and wherein the reservoir layer comprises a second configuration of fluid reservoirs on a second side of the microfluidic chip, wherein the first side is opposite the second side, and wherein the fluid reservoirs on the second side comprise a first fluid reservoir and a second fluid reservoir.
12. The microfluidic chip of claim 11, wherein the reservoir layer comprises a gasket, wherein the first fluid reservoir is defined by a first void in a gasket, and wherein the second fluid reservoir is defined by a second void in the gasket.
13. The microfluidic chip of claim 12, wherein the gasket comprises a geometric shape formed by the first void and the second void, and wherein the geometric shape is selected from the group consisting of a Greek spiral and a serpentine shape.
14. The microfluidic chip of claim 11, wherein at least one of the first configuration or second configuration comprises a serpentine configuration.
15. The microfluidic chip of claim 11, wherein at least one of the first configuration or the second configuration comprises a Greek spiral configuration.
16. The microfluidic chip of claim 11, wherein the first configuration and second configuration comprise a serpentine configuration.
17. The microfluidic chip of claim 11, wherein the first configuration and the second configuration comprise a Greek spiral configuration.
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